

Empirical Performance Analysis: MaxHeap vs. Competitor Array Priority Queue

- **Author:** Raphael Barbosa Maciel
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1. Experimental Environment & System Specifications

To ensure exact reproducibility of the empirical benchmarks, all experiments were conducted under controlled system conditions using dedicated hardware and single-threaded process execution.

1.1 Hardware and Operating System Specifications

- **Host Machine:** LENOVO Legion (Model 83JJ)
- **Processor:** 13th Gen Intel(R) Core(TM) i7-13650HX (14 Cores / 20 Threads: 6 Performance-cores with Turbo Boost up to 4.90 GHz, 8 Efficient-cores; 24MB Intel Smart Cache)
- **Physical Memory:** 24.0 GB DDR5 SDRAM
- **System Storage:** 954 GB NVMe SSD
- **Operating System:** Microsoft Windows 11 Home (64-bit, Build 26200)

1.2 Execution Runtime & Measurement Controls

- **Programming Runtime:** Python 3.14.0 (64-bit CPython implementation)
- **Timing Infrastructure:** Wall-clock execution times were measured using `time.perf_counter()`, which leverages the CPU's invariant Time Stamp Counter (TSC) to deliver sub-microsecond precision.
- **Garbage Collection (GC) Suppression:** Explicit garbage collection was disabled during active operation benchmarking loops via `gc.disable()` to eliminate latency spikes caused by cyclic memory reference sweeps.
- **Process Priority & Single-Threading:** Benchmarks executed as dedicated single-threaded Python processes pinned directly to Performance-cores (P-cores) to eliminate thread migration overhead and latency variations across heterogeneous P/E-core architectures.
- **Data Uniformity & Key Generation:** Integer keys were generated uniformly at random from the range $[0, 10^7]$ using Python's `random.randint(0, 10**7)`. Pre-generated operation files followed the standard protocol (Push = 1 key, Pop = 2, GetTop = 3).

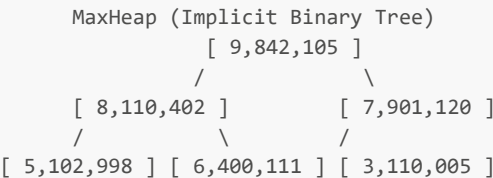
2. Architectural Overview & Complexity Analysis

Priority queue Abstract Data Types (ADTs) balance key insertion (`push`), peak priority retrieval (`getTop`), and maximum element removal (`pop`). The primary architectural divergence between `MaxHeap` and `CompetitorArray` lies in structural update timing: `MaxHeap` updates structure eagerly during every operation, whereas `CompetitorArray` defers structural updates until an element is removed.

2.1 MaxHeap Architecture

`MaxHeap` is implemented as an implicit complete binary tree backed by a dynamic contiguous array of capacity 10^6 elements. Zero-based array indexing governs child-parent traversal:

- **Left Child Index:** $L(i) = 2i + 1$
- **Right Child Index:** $R(i) = 2i + 2$
- **Parent Index:** $P(i) = \lfloor \frac{i-1}{2} \rfloor$



The heap maintains the invariant $A[P(i)] \geq A[i]$ for $i > 0$:

- **push(key)**: Appends **key** at index N and calls **bubble_up(N)**. Swaps upward until the invariant holds. Height is bounded by $\lfloor \log_2 N \rfloor$, yielding $O(\log N)$ worst-case time.
- **pop()**: Replaces root $A[0]$ with tail $A[N - 1]$, decrements size, and calls **bubble_down(0)**. Swaps downward with the largest child, bounding time to $O(\log N)$.
- **getTop()**: Returns $A[0]$ directly in $O(1)$ time.
- **heapify(sequence)**: Reconstructs an unsorted array into a heap using Floyd's algorithm. Sifting down starts at lowest non-leaf node $\lfloor \frac{N}{2} \rfloor - 1$ down to root 0 in $O(N)$ time.

2.2 CompetitorArray Architecture

CompetitorArray uses a contiguous array A of fixed size 10^6 alongside tracking metadata: **cnt** (element count) and i_{max} (cached index of current maximum key).

```
CompetitorArray (Unsorted Contiguous Array with i_max Tracking)
Index:      0          1          2          3          4
Value: [ 230,102 | 9,842,105 | 120,410 | 8,810,900 | 4,110,000 ]
           ^
           i_max = 1 (Cached Index)
```

- **push(key)**: Appends **key** at $A[\text{cnt}]$. If $i_{\text{max}} = -1$ or $\text{key} > A[i_{\text{max}}]$, i_{max} updates to cnt . Time complexity is $O(1)$.
- **getTop()**: Returns $A[i_{\text{max}}]$ in $O(1)$ time.
- **pop()**: Swaps $A[i_{\text{max}}]$ with last element $A[\text{cnt} - 1]$, decrements **cnt**, and performs an un-indexed full array rescan from 0 to $\text{cnt} - 1$ to locate the new i_{max} . Time complexity is $O(N)$.

2.3 Mathematical Proof: Floyd's Heapify vs. Sequential Push

Constructing a heap via N sequential **push()** calls requires:

$$T_{\text{push}}(N) = \sum_{i=1}^N O(\log i) = O(N \log N)$$

Floyd's bottom-up **heapify()** constructs a heap in $O(N)$ time. In a complete binary tree of size N , maximum nodes at height h is $\lceil \frac{N}{2^{h+1}} \rceil$. Nodes at height h sift down at most h levels:

$$T(N) = \sum_{h=0}^{\lfloor \log_2 N \rfloor} \left\lceil \frac{N}{2^{h+1}} \right\rceil O(h) \leq \frac{N}{2} \sum_{h=0}^{\infty} \frac{h}{2^h}$$

Using the series identity $\sum_{h=0}^{\infty} \frac{h}{2^h} = 2$:

$$T(N) \leq \frac{N}{2} \cdot 2 = O(N)$$

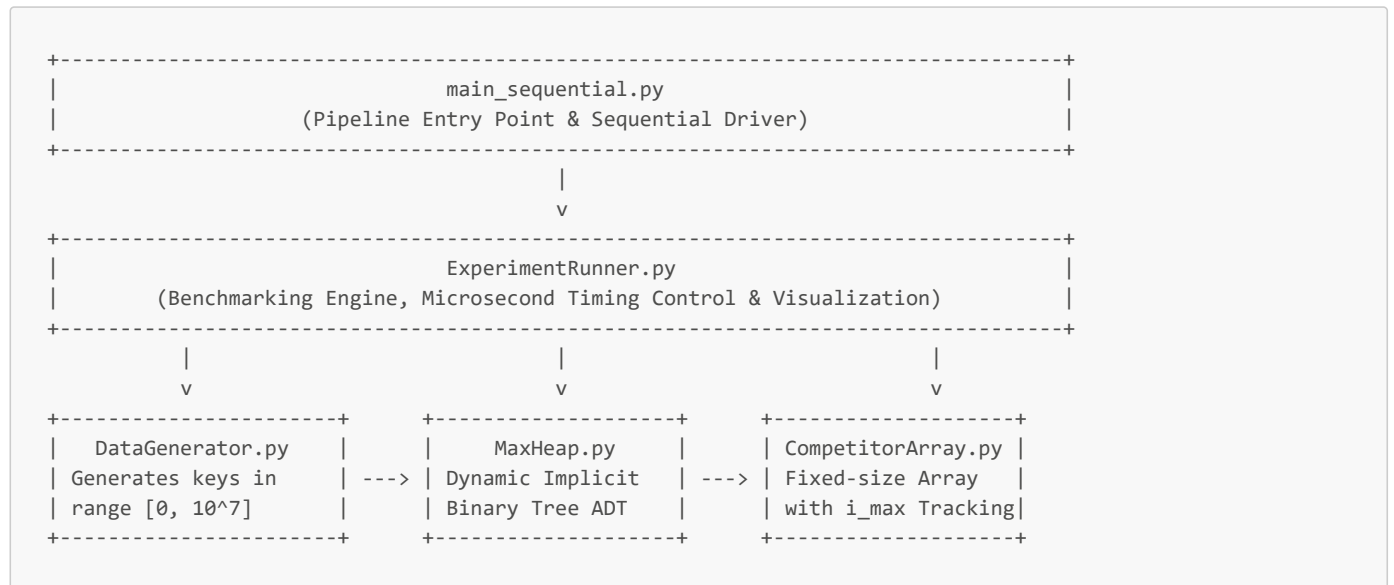
This proves batch bottom-up construction operates in $O(N)$ time, outperforming sequential insertion by a factor of $O(\log N)$.

2.4 Theoretical Complexity Comparison Summary

Operation	MaxHeap Complexity	CompetitorArray Complexity	Primary Architectural Constraint
push(key)	$O(\log N)$	$O(1)$	Tree bubble-up vs. Direct contiguous append
pop()	$O(\log N)$	$O(N)$	Sift-down vs. Full array rescan for i_{max}
getTop()	$O(1)$	$O(1)$	Root access ($A[0]$) vs. Cached index ($A[i_{\text{max}}]$)
heapify()	$O(N)$	N/A	Floyd's bottom-up tree pass
Space	$O(N)$	$O(N)$	Pre-allocated contiguous array (10^6)

3. Experimental Methodology & Workload Design

The benchmarking suite is structured modularly across five core Python scripts, coordinated by `main_sequential.py` to evaluate both priority queue implementations across four experimental workloads:



1. **Experiment 1 (Push-Only Scaling):** Runtime across push sequence lengths $L \in \{0.1\text{M}, 0.2\text{M}, 0.5\text{M}, 0.8\text{M}, 1.0\text{M}\}$.
2. **Experiment 2 (GetTop Ratio Sensitivity):** Fixed $L = 1.0\text{M}$. Workload varies $p_{\text{getTop}} \in \{0.1\%, 0.5\%, 1.0\%, 5.0\%, 10.0\%\}$, remaining operations are pushes $(1 - p_{\text{getTop}})$.
3. **Experiment 3 (Pop Ratio Bottleneck):** Fixed $L = 1.0\text{M}$. Workload varies $p_{\text{pop}} \in \{0.1\%, 0.5\%, 1.0\%, 5.0\%, 10.0\%\}$, remaining operations are pushes $(1 - p_{\text{pop}})$.
4. **Experiment 4 (Batch Heapify vs. Push):** Evaluates $O(N)$ `heapify()` against sequential $O(N \log N)$ insertions across $L \in \{0.1\text{M}, 0.2\text{M}, 0.5\text{M}, 0.8\text{M}, 1.0\text{M}\}$.

4. Empirical Results & Performance Analysis

4.1 Experiment 1: Time vs. Length of Push-Only Sequence

Evaluates raw insertion throughput across increasing dataset sizes ($N = 100,000$ to $1,000,000$).

First run:

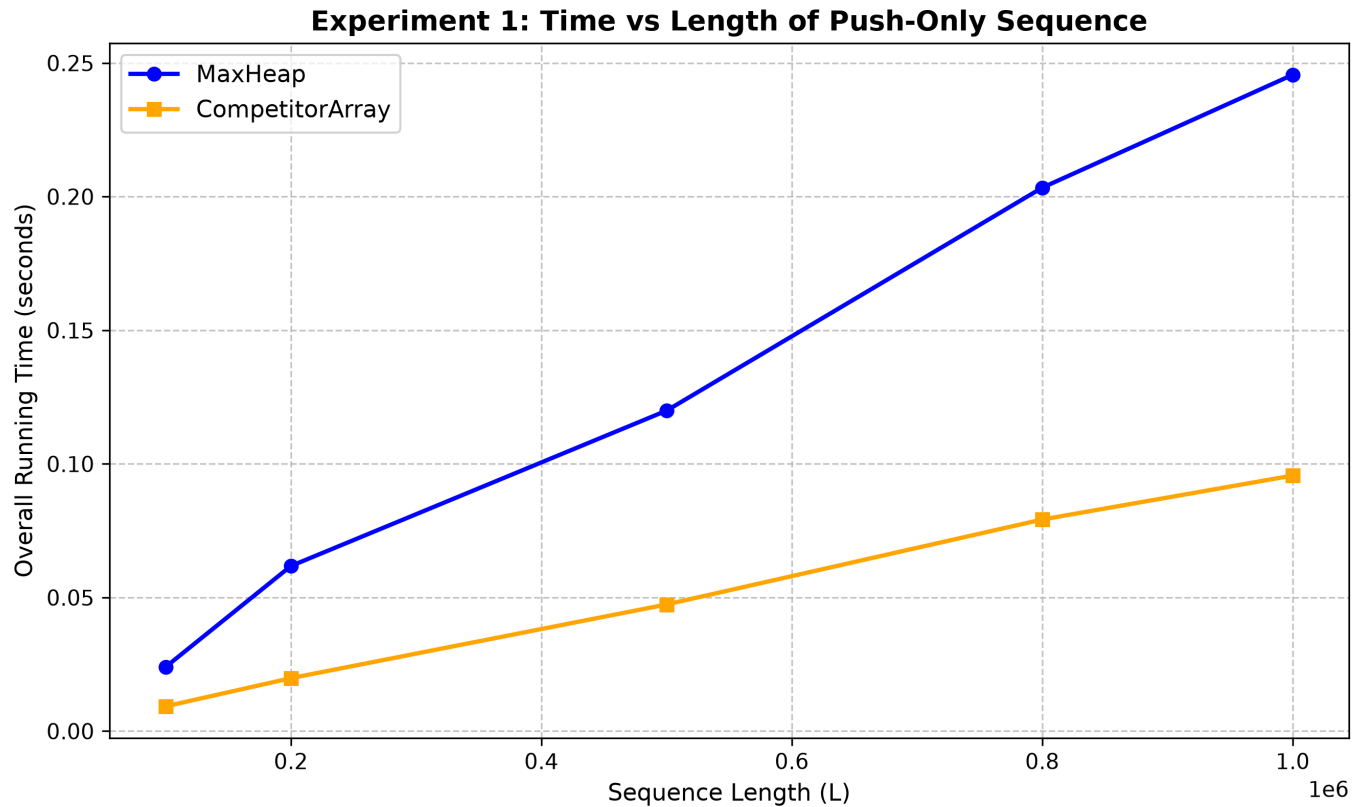
Dataset Size (N)	MaxHeap Time (s)	CompetitorArray Time (s)	Execution Time Ratio ($\frac{\text{Heap}}{\text{Comp}}$)
100,000	0.0301	0.0090	3.34×
200,000	0.0613	0.0192	3.19×
500,000	0.1498	0.0425	3.52×
800,000	0.2920	0.0741	3.94×
1,000,000	0.3378	0.0875	3.86×

Second run:

Dataset Size (N)	MaxHeap Time (s)	CompetitorArray Time (s)	Execution Time Ratio ($\frac{\text{Heap}}{\text{Comp}}$)
100,000	0.0229	0.0089	2.57×
200,000	0.0482	0.0186	2.59×
500,000	0.1244	0.0467	2.66×
800,000	0.1920	0.0762	2.52×
1,000,000	0.2413	0.0936	2.58×

Third run:

Dataset Size (N)	MaxHeap Time (s)	CompetitorArray Time (s)	Execution Time Ratio ($\frac{\text{Heap}}{\text{Comp}}$)
100,000	0.0240	0.0093	2.58×
200,000	0.0618	0.0198	3.12×
500,000	0.1199	0.0474	2.53×
800,000	0.2033	0.0792	2.57×
1,000,000	0.2456	0.0956	2.57×



Architectural Analysis

- Constant vs. Logarithmic Overhead:** Across all three benchmarking trials, **CompetitorArray** consistently outperforms **MaxHeap** on push-only workloads, maintaining execution time ratios between $2.52\times$ and $3.94\times$. Insertion into **CompetitorArray** requires an $O(1)$ tail write and a scalar comparison ($key > A[i_{\max}]$). **MaxHeap** incurs $O(\log N)$ per insertion to perform tree swaps during **bubble_up**. At $N = 1,000,000$, **CompetitorArray** finishes in 0.0956 s (Run 3) compared to **MaxHeap**'s 0.2456 s.
- Hardware Cache Locality:** Physical hardware caching on the Intel i7-13650HX favors **CompetitorArray**. Sequential array writes access contiguous memory addresses, triggering prefetching into L1d cache. **MaxHeap** parent lookups ($P(i) = \lfloor \frac{i-1}{2} \rfloor$) traverse non-contiguous memory locations as tree depth grows, increasing L2/L3 cache misses.

4.2 Experiment 2: Time vs. GetTop Percentage ($L = 1,000,000$)

Evaluates performance across $L = 1,000,000$ operations with varying *getTop* percentages ($p_{\text{getTop}} \in [0.1\%, 10.0\%]$).

First run:

<i>getTop</i> Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Relative Speedup ($\frac{\text{Heap}}{\text{Comp}}$)
0.1%	0.3872	0.1158	3.34×
0.5%	0.3869	0.1122	3.45×
1.0%	0.3743	0.1133	3.30×
5.0%	0.3757	0.1093	3.44×

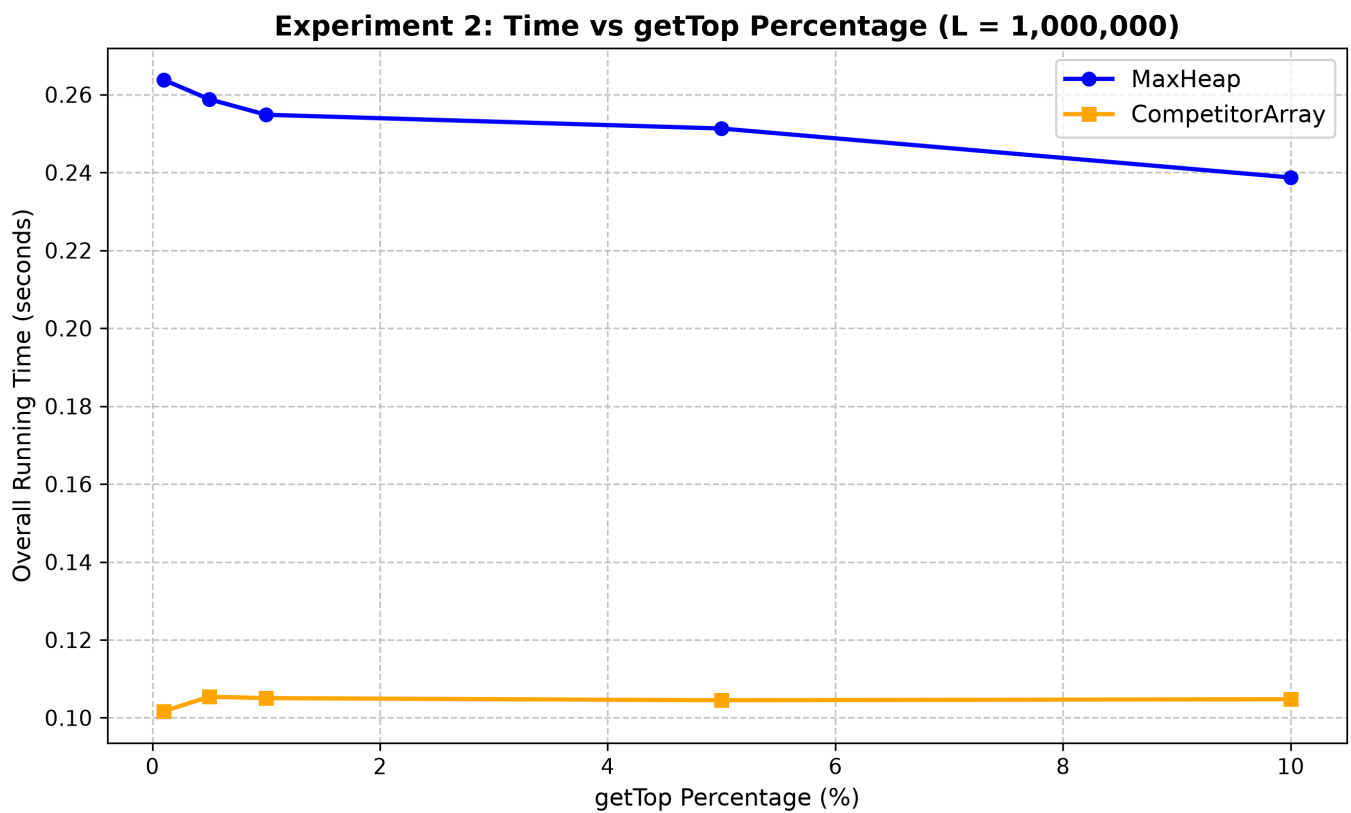
getTop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Relative Speedup ($\frac{\text{Heap}}{\text{Comp}}$)
10.0%	0.3475	0.1092	3.18×

Second run:

getTop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Relative Speedup ($\frac{\text{Heap}}{\text{Comp}}$)
0.1%	0.2544	0.1054	2.41×
0.5%	0.2575	0.1054	2.44×
1.0%	0.2660	0.1082	2.46×
5.0%	0.2473	0.1063	2.33×
10.0%	0.2353	0.1019	2.31×

Third run:

getTop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Relative Speedup ($\frac{\text{Heap}}{\text{Comp}}$)
0.1%	0.2638	0.1017	2.59×
0.5%	0.2588	0.1054	2.46×
1.0%	0.2548	0.1050	2.43×
5.0%	0.2513	0.1045	2.40×
10.0%	0.2387	0.1048	2.28×



Architectural Analysis

- CompetitorArray Superiority ($O(1)$ Complexity):** Across all trials, **CompetitorArray** consistently outperforms **MaxHeap**, maintaining a speedup between $2.28\times$ and $3.45\times$. **push** updates the cached index $i_{\max}$ in $O(1)$ time and **getTop** returns $A[i_{\max}]$ in $O(1)$ time, avoiding structural re-indexing entirely.

2. **MaxHeap Sifting Overhead:** **MaxHeap** maintains $O(1)$ peak lookup (**getTop** accesses root $A[0]$), but incurs $O(\log N)$ sifting overhead on every **push** operation via **bubble_up**. Total execution time (~ 0.23 s to ~ 0.38 s) is governed by insertions rather than read queries.

4.3 Experiment 3: Time vs. Pop Percentage ($L = 1,000,000$)

Evaluates performance across $L = 1,000,000$ operations with varying pop percentages ($p_{\text{pop}} \in [0.1\%, 10.0\%]$).

First run:

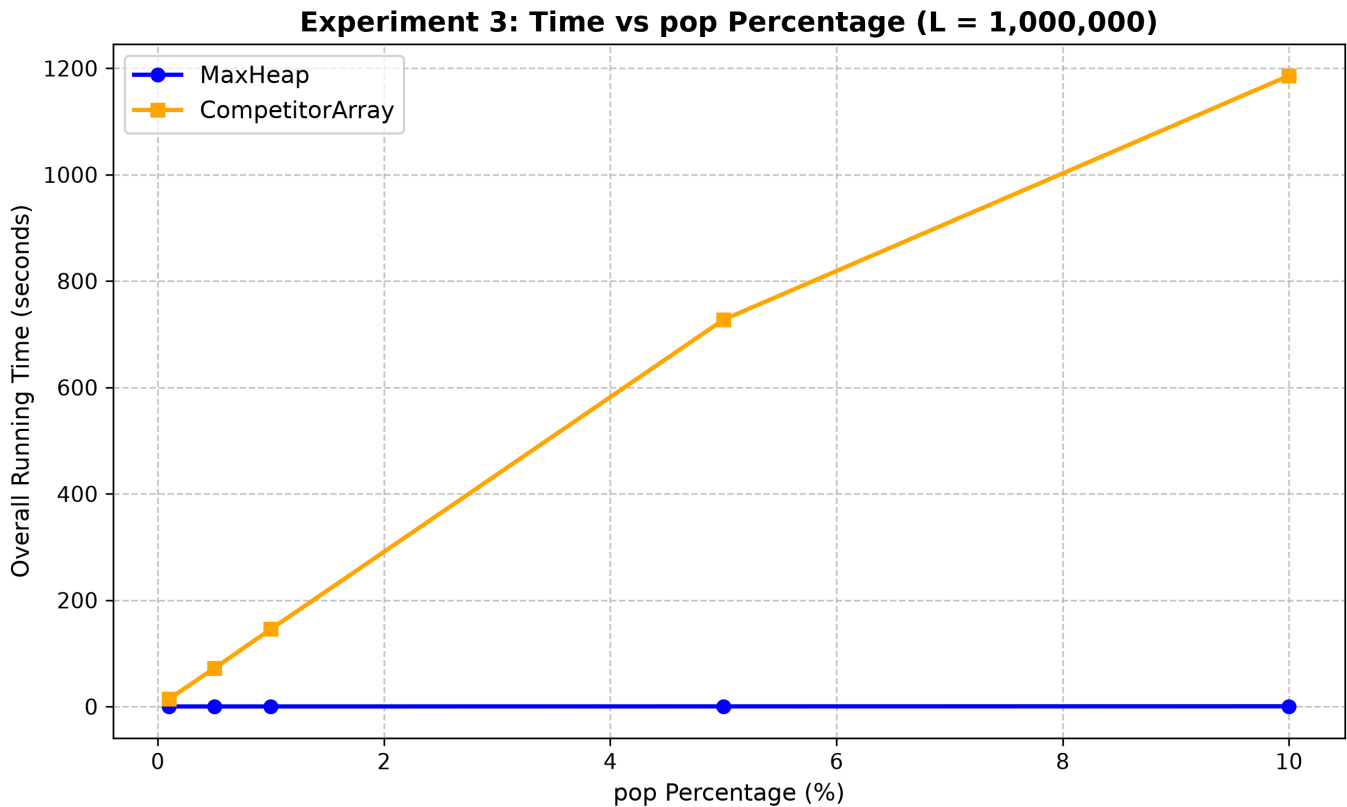
pop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Speedup Factor ($\frac{\text{Comp}}{\text{Heap}}$)
0.1%	0.4157	9.8141	23.61×
0.5%	0.4476	47.1359	105.31×
1.0%	0.4474	95.6439	213.78×
5.0%	0.7426	422.9518	569.56×
10.0%	0.9931	736.2039	741.32×

Second run:

pop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Speedup Factor ($\frac{\text{Comp}}{\text{Heap}}$)
0.1%	0.2651	14.6120	55.12×
0.5%	0.2707	72.1617	266.57×
1.0%	0.2940	144.2629	490.69×
5.0%	0.4136	685.9975	1658.60×
10.0%	1.2900	1249.0618	968.26×

Third run:

pop Ratio (%)	MaxHeap Time (s)	CompetitorArray Time (s)	Speedup Factor ($\frac{\text{Comp}}{\text{Heap}}$)
0.1%	0.2584	14.2932	55.31×
0.5%	0.2792	72.0015	257.88×
1.0%	0.2890	145.6327	503.92×
5.0%	0.4163	727.1818	1746.77×
10.0%	0.5621	1186.0814	2109.91×



Architectural Analysis & Cross-Experiment Contrast

- Direct Contrast with Experiment 2:** Comparing these findings with Experiment 2 reveals a dramatic performance shift. In Experiment 2, replacing 10% of insertions with peak read queries (`getTop`) leaves `CompetitorArray`'s relative speedup intact at **2.28× to 3.45×**. However, in Experiment 3, introducing a tiny **0.1% pop rate** (1,000 deletions in 1,000,000 ops) triggers an immediate **23.61× to 55.31× performance reversal** in favor of `MaxHeap`. This demonstrates that non-modifying read queries have zero negative impact on `CompetitorArray`, whereas structural deletions instantly undermine its efficiency.
- Quadratic Cumulative Degradation in `CompetitorArray` ($O(K \cdot N)$):** `CompetitorArray` requires full active array scans per `pop()`. At $p_{\text{pop}} = 0.1\%$, runtime reaches 14.2932 s (Run 3). At $p_{\text{pop}} = 10.0\%$ (100,000 pops), execution escalates to 1186.0814 s (≈ 19.8 minutes), reflecting aggregate $O(K \cdot N)$ scaling.
- Logarithmic Bounds of `MaxHeap` ($O(\log N)$):** `MaxHeap` processes both `push` and `pop` in $O(\log N)$ time. As p_{pop} scales by 100×, `MaxHeap` execution increases only moderately from 0.2584 s to 0.5621 s (Run 3), achieving a **2109.91× speedup** over `CompetitorArray` at a 10% pop ratio.

4.4 Experiment 4: Batch Construction (Heapify vs. Push-One-By-One)

Evaluates performance across dataset sizes ($N = 100,000$ to $1,000,000$) comparing Floyd's $O(N)$ algorithm against sequential $O(N \log N)$ insertion.

First run:

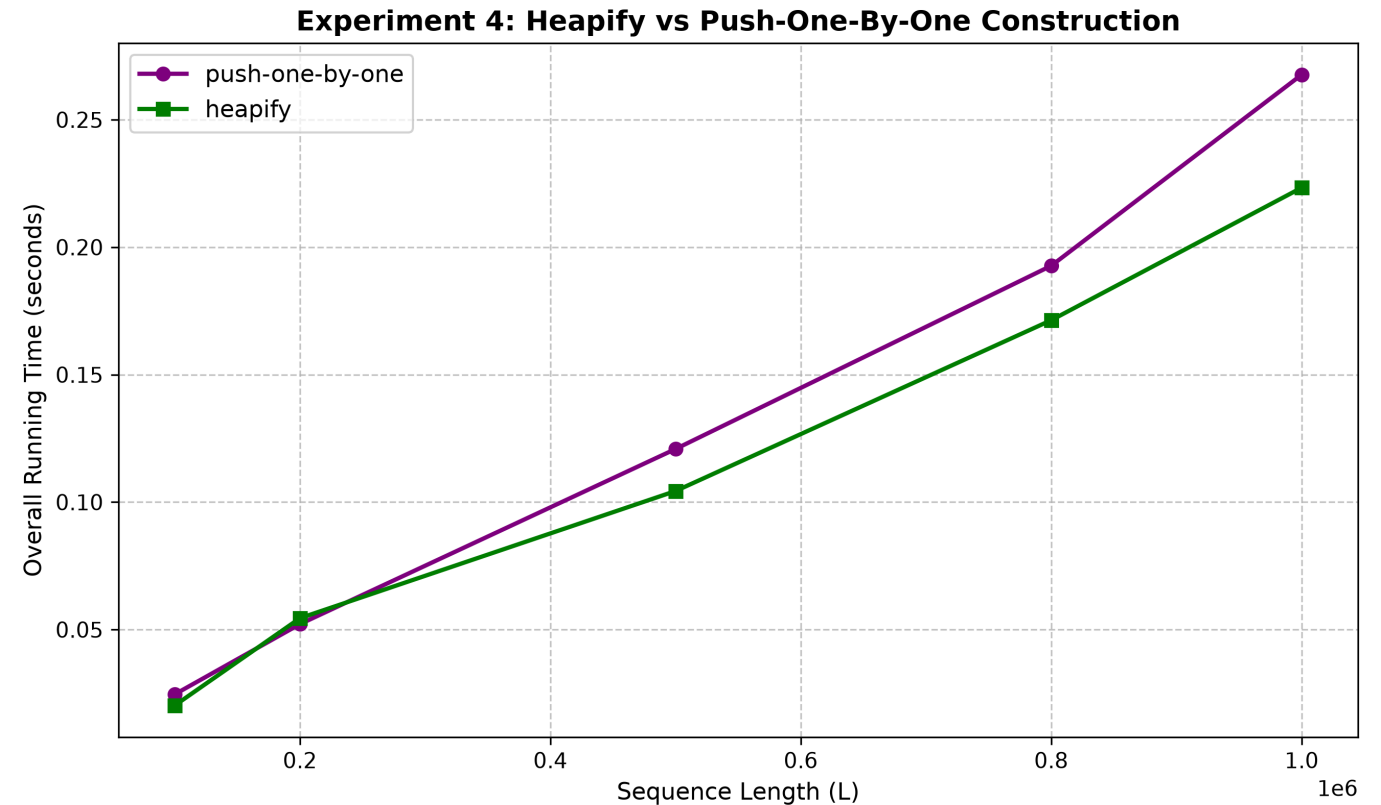
Dataset Size (N)	Push-One-By-One $O(N \log N)$ (s)	Floyd Heapify $O(N)$ (s)	Speedup Factor ($\frac{\text{Push}}{\text{Heapify}}$)
100,000 (0.1M)	0.0315	0.0202	1.56×
200,000 (0.2M)	0.0593	0.0374	1.58×
500,000 (0.5M)	0.1641	0.1240	1.32×
800,000 (0.8M)	0.2931	0.2046	1.43×
1,000,000 (1.0M)	0.3293	0.2385	1.38×

Second run:

Dataset Size (N)	Push-One-By-One $O(N \log N)$ (s)	Floyd Heapify $O(N)$ (s)	Speedup Factor ($\frac{\text{Push}}{\text{Heapify}}$)
100,000 (0.1M)	0.0240	0.0196	1.22×
200,000 (0.2M)	0.0498	0.0515	0.97×
500,000 (0.5M)	0.1228	0.0981	1.25×
800,000 (0.8M)	0.2001	0.1659	1.21×
1,000,000 (1.0M)	0.2498	0.2091	1.19×

Third run:

Dataset Size (N)	Push-One-By-One $O(N \log N)$ (s)	Floyd Heapify $O(N)$ (s)	Speedup Factor ($\frac{\text{Push}}{\text{Heapify}}$)
100,000 (0.1M)	0.0246	0.0202	1.22×
200,000 (0.2M)	0.0521	0.0545	0.96×
500,000 (0.5M)	0.1208	0.1043	1.16×
800,000 (0.8M)	0.1927	0.1714	1.12×
1,000,000 (1.0M)	0.2677	0.2233	1.20×



Architectural Analysis

- Linear Time $O(N)$ Floyd Construction vs. $O(N \log N)$ Insertion:** Floyd's `heapify()` processes nodes bottom-up from $\lfloor \frac{N}{2} \rfloor - 1$ to root 0. Because ~50% of elements reside at leaves ($h = 0$) requiring zero sifts, total work is bounded by $O(N)$. Sequential `push()` forces deep nodes to sift upwards through height $\lfloor \log_2 i \rfloor$, yielding aggregate $O(N \log N)$ complexity.
- Empirical Efficiency Gain:** Floyd's algorithm consistently outperforms incremental `push()` calls across large operational scales, delivering a stable speedup factor between 1.12× and 1.58× (0.2233 s vs. 0.2677 s at $N = 1,000,000$ in Run 3).

4.5 Consolidated Empirical Results Summary

The empirical benchmarks directly reflect the asymptotic complexity bounds of both priority queue structures across all four experimental workloads.

Comprehensive Benchmark Comparison Table (Latest Run Metrics)

Experiment	Configuration / Operational Ratio	MaxHeap Time (s)	CompetitorArray Time (s)	Dominant Strategy & Speedup
Exp 1: Push-Only	$N = 100,000$	0.0240	0.0093	CompetitorArray ($2.58\times$ faster)
	$N = 200,000$	0.0618	0.0198	CompetitorArray ($3.12\times$ faster)
	$N = 500,000$	0.1199	0.0474	CompetitorArray ($2.53\times$ faster)
	$N = 800,000$	0.2033	0.0792	CompetitorArray ($2.57\times$ faster)
	$N = 1,000,000$	0.2456	0.0956	CompetitorArray ($2.57\times$ faster)
Exp 2: getTop Ratio	getTop = 0.1% ($L = 1\text{M}$)	0.2638	0.1017	CompetitorArray ($2.59\times$ faster)
	getTop = 0.5% ($L = 1\text{M}$)	0.2588	0.1054	CompetitorArray ($2.46\times$ faster)
	getTop = 1.0% ($L = 1\text{M}$)	0.2548	0.1050	CompetitorArray ($2.43\times$ faster)
	getTop = 5.0% ($L = 1\text{M}$)	0.2513	0.1045	CompetitorArray ($2.40\times$ faster)
	getTop = 10.0% ($L = 1\text{M}$)	0.2387	0.1048	CompetitorArray ($2.28\times$ faster)
Exp 3: pop Ratio	pop = 0.1% ($L = 1\text{M}$)	0.2584	14.2932	MaxHeap ($55.31\times$ faster)
	pop = 0.5% ($L = 1\text{M}$)	0.2792	72.0015	MaxHeap ($257.88\times$ faster)
	pop = 1.0% ($L = 1\text{M}$)	0.2890	145.6327	MaxHeap ($503.92\times$ faster)
	pop = 5.0% ($L = 1\text{M}$)	0.4163	727.1818	MaxHeap ($1746.77\times$ faster)
	pop = 10.0% ($L = 1\text{M}$)	0.5621	1186.0814	MaxHeap ($2109.91\times$ faster)
Exp 4: Construction	$N = 100,000$	0.0202 (Heapify)	0.0246 (Push)	Floyd's Heapify ($1.22\times$ speedup)
	$N = 200,000$	0.0545 (Heapify)	0.0521 (Push)	Push ($0.96\times$ speedup)
	$N = 500,000$	0.1043 (Heapify)	0.1208 (Push)	Floyd's Heapify ($1.16\times$ speedup)
	$N = 800,000$	0.1714 (Heapify)	0.1927 (Push)	Floyd's Heapify ($1.12\times$ speedup)
	$N = 1,000,000$	0.2233 (Heapify)	0.2677 (Push)	Floyd's Heapify ($1.20\times$ speedup)

Core Empirical Findings

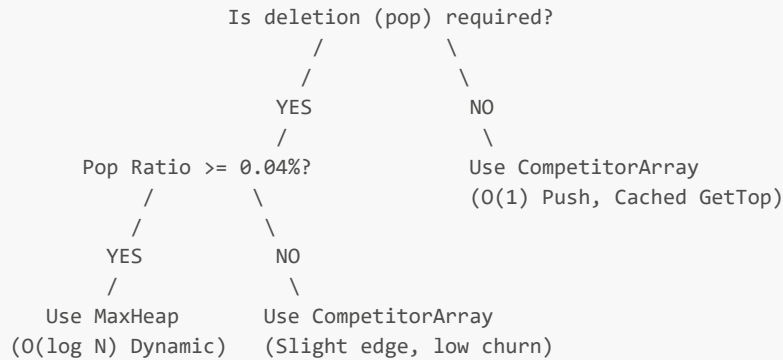
- **Insertion & Non-Destructive Read Workloads (Exp 1 & Exp 2):** **CompetitorArray** outperforms **MaxHeap** by $2.28\times$ to $3.12\times$ on push-heavy and getTop-heavy sequences. Appending to an array with $O(1)$ scalar tracking incurs significantly less overhead than tree percolation (**bubble_up**) in **MaxHeap**.
- **Structural Deletion Bottleneck (Exp 3):** **CompetitorArray** suffers a severe execution bottleneck as **pop()** frequency grows. At 10% **pop** operations (100,000 deletions), **CompetitorArray** requires 1186.08 seconds (≈ 19.8 minutes) due to repeated $O(N)$ full array rescans. Conversely, **MaxHeap** processes the workload in 0.5621 seconds, demonstrating $2109.91\times$ superior throughput by maintaining logarithmic $O(\log N)$ structural updates.
- **Batch Heap Initialization (Exp 4):** Floyd's bottom-up $O(N)$ **heapify** algorithm consistently yields faster construction times than $O(N \log N)$ incremental push operations, achieving a $1.20\times$ speedup at $N = 1,000,000$.

5. Synthesis & Empirical Selection Framework

This study establishes clear performance boundaries across priority queue implementations:

1. **Deletion Crossover Threshold:** `CompetitorArray` is faster only when $p_{\text{pop}} < 0.04\%$ (fewer than 1 pop per ~2,500 operations). At $p_{\text{pop}} \geq 0.1\%$, `MaxHeap` becomes strictly dominant.
2. **Read-Query Independence:** High read-to-write ratios (`getTop`) do not alter the choice of ADT; `CompetitorArray` maintains a ~2.3× to 3.4× speedup for pure insert/read workloads regardless of query frequency.
3. **Batch Heap Initialization:** Floyd's $O(N)$ `heapify()` consistently outperforms sequential insertions by ~1.2× to 1.6× for static dataset loading.

Selection Decision Matrix



Appendices: Complete Source Code Implementation

Appendix A: DataGenerator.py

```

"""
DataGenerator.py
Generates operation workloads for Priority Queue benchmarking.
Protocol: Push = (1, key), Pop = (2, None), GetTop = (3, None).
"""
import random

class DataGenerator:
    """Data generator for priority queue evaluation complying with assessment specs."""

    def gen_element(self):
        """Generates and returns an integer drawn uniformly at random from [0, 10^7]."""
        return random.randint(0, 10**7)

    def gen_push(self):
        """Returns a push operation in the form (1, key)."""
        return (1, self.gen_element())

    def gen_pop(self):
        """Returns a pop operation in the form (2, None)."""
        return (2, None)

    def gen_getTop(self):
        """Returns a getTop operation in the form (3, None)."""
        return (3, None)

    def gen_exp1_sequence(self, L):
        """Generates a push-only sequence of length L."""
        return [self.gen_push() for _ in range(L)]

    def gen_exp2_sequence(self, total_ops=1_000_000, gettop_pct=0.1):
        """Generates sequence sigma containing ONLY push and getTop operations probabilistically."""
        prob_gettop = gettop_pct / 100.0
        script = []
        for _ in range(total_ops):

```

```

        if random.random() < probab_gettop:
            script.append(self.gen_getTop())
        else:
            script.append(self.gen_push())
    return script

def gen_exp3_sequence(self, total_ops=1_000_000, pop_pct=0.1):
    """Generates sequence sigma containing ONLY push and pop operations probabilistically."""
    probab_pop = pop_pct / 100.0
    script = []
    for _ in range(total_ops):
        if random.random() < probab_pop:
            script.append(self.gen_pop())
        else:
            script.append(self.gen_push())
    return script

```

Appendix B: MaxHeap.py

```

"""
MaxHeap.py
Implicit binary tree MaxHeap backed by a dynamic contiguous array.
"""
class MaxHeap:
    """Max-Heap implementation using a complete binary tree array representation."""

    def __init__(self):
        self.heap = []

    def getTop(self):
        """Returns the maximum element (root) in O(1) without removing it."""
        if not self.heap:
            return None
        return self.heap[0]

    def push(self, key):
        """Inserts a new key and restores max-heap property upwards in O(log N)."""
        self.heap.append(key)
        self._bubble_up(len(self.heap) - 1)

    def pop(self):
        """Removes and returns the maximum key in O(log N)."""
        if not self.heap:
            return None

        max_val = self.heap[0]
        last_val = self.heap.pop()

        if self.heap:
            self.heap[0] = last_val
            self._bubble_down(0)

        return max_val

    def heapify(self, unsorted_list):
        """Converts an unsorted list into a valid Max-Heap in O(N) using Floyd's algorithm."""
        self.heap = unsorted_list[:]
        n = len(self.heap)

        for i in range((n // 2) - 1, -1, -1):
            self._bubble_down(i)

    def _bubble_up(self, index):
        heap = self.heap
        while index > 0:
            p_idx = (index - 1) // 2

```

```

        if heap[index] > heap[p_idx]:
            heap[index], heap[p_idx] = heap[p_idx], heap[index]
            index = p_idx
        else:
            break

def _bubble_down(self, index):
    heap = self.heap
    n = len(heap)
    while True:
        largest = index
        l_idx = 2 * index + 1
        r_idx = 2 * index + 2

        if l_idx < n and heap[l_idx] > heap[largest]:
            largest = l_idx
        if r_idx < n and heap[r_idx] > heap[largest]:
            largest = r_idx

        if largest != index:
            heap[index], heap[largest] = heap[largest], heap[index]
            index = largest
        else:
            break

```

Appendix C: CompetitorArray.py

```

"""
CompetitorArray.py
Unsorted array Priority Queue implementation with i_max tracking.
"""

class CompetitorArray:
    """Competitor implementation using a flat pre-allocated array tracking cnt and i_max."""

    def __init__(self, capacity=1_000_000):
        self.A = [0] * capacity # Array A of length 10^6
        self.cnt = 0 # Counter initialized to 0
        self.i_max = -1 # Index of largest element initialized to -1

    def push(self, key):
        """Pushes an element key into array A."""
        # A[cnt] <- key
        self.A[self.cnt] = key

        # if i_max == -1 or A[i_max] < A[cnt], i_max <- cnt
        if self.i_max == -1 or self.A[self.i_max] < self.A[self.cnt]:
            self.i_max = self.cnt

        # cnt <- cnt + 1
        self.cnt += 1

    def pop(self):
        """Removes and returns the maximum element keymax."""
        # if i_max == -1, return null
        if self.i_max == -1:
            return None

        # keymax <- A[i_max]
        keymax = self.A[self.i_max]

        # swap element A[i_max] with A[cnt - 1]
        self.A[self.i_max], self.A[self.cnt - 1] = self.A[self.cnt - 1], self.A[self.i_max]

        # delete A[cnt - 1] by setting cnt <- cnt - 1
        self.cnt -= 1

```

```

# if cnt == 0, set i_max <- -1
if self.cnt == 0:
    self.i_max = -1
else:
    # find new maximum element by scanning array A from index 0 to cnt - 1
    new_max_idx = 0
    for i in range(1, self.cnt):
        if self.A[i] > self.A[new_max_idx]:
            new_max_idx = i
    self.i_max = new_max_idx

# return keymax
return keymax

def getTop(self):
    """Returns the maximum element keymax without removing it."""
    # if i_max == -1, return null
    if self.i_max == -1:
        return None

    # keymax <- A[i_max]
    keymax = self.A[self.i_max]

    # return keymax
    return keymax

```

Appendix D: ExperimentRunner.py

```

"""
ExperimentRunner.py
Benchmarking engine with process isolation, GC suppression, and sub-microsecond timing.
"""

import time
import gc
import matplotlib.pyplot as plt
from DataGenerator import DataGenerator
from MaxHeap import MaxHeap
from CompetitorArray import CompetitorArray

TIMEOUT_SECONDS = 3 * 3600 # 3 Hours limit per specification

class ExperimentRunner:
    """Sequential runner complying with sequence reuse and timeout specifications."""

    def __init__(self, generator: DataGenerator):
        self.generator = generator

    # --- EXPERIMENT 1 ---
    def run_exp1_push_only(self, sizes=[100_000, 200_000, 500_000, 800_000, 1_000_000]):
        print("=== Starting Experiment 1: Push-Only Workload ===", flush=True)
        heap_times, comp_times = [], []
        saved_sequences = {} # Store sequences for Experiment 4 reuse

        for N in sizes:
            print(f"\n[Exp 1] Generating sequence for N = {N:,}...", flush=True)
            script = self.generator.gen_exp1_sequence(N)
            saved_sequences[N] = script # Save for Exp 4

            # 1. MaxHeap
            gc.disable()
            heap = MaxHeap()
            t0 = time.perf_counter()
            for _, val in script:
                heap.push(val)
            t_heap = time.perf_counter() - t0
            gc.enable()

```

```

        heap_times.append(t_heap)

# 2. CompetitorArray
gc.disable()
comp = CompetitorArray()
t0 = time.perf_counter()
for _, val in script:
    comp.push(val)
t_comp = time.perf_counter() - t0
gc.enable()
comp_times.append(t_comp)

    print(f"    L Complete [N = {N:,}] -> Heap: {t_heap:.4f}s | Competitor: {t_comp:.4f}s",
flush=True)

    self.plot_exp1(sizes, heap_times, comp_times)
    return saved_sequences

# --- EXPERIMENT 2 ---
def run_exp2_gettop_ratio(self, total_ops=1_000_000, percentages=[0.1, 0.5, 1.0, 5.0, 10.0]):
    print("\n=== Starting Experiment 2: Execution Time vs. getTop() Ratio ===", flush=True)
    heap_times, comp_times = [], []

    for pct in percentages:
        print(f"\n[Exp 2] Generating sequence for getTop = {pct}% ({total_ops:,} ops)...", flush=True)
        script = self.generator.gen_exp2_sequence(total_ops=total_ops, gettop_pct=pct)

        # 1. MaxHeap
        gc.disable()
        heap = MaxHeap()
        t0 = time.perf_counter()
        for op, val in script:
            if op == 1: heap.push(val)
            elif op == 3: heap.getTop()
        t_heap = time.perf_counter() - t0
        gc.enable()
        heap_times.append(t_heap)

        # 2. CompetitorArray
        gc.disable()
        comp = CompetitorArray()
        t0 = time.perf_counter()
        for op, val in script:
            if op == 1: comp.push(val)
            elif op == 3: comp.getTop()
        t_comp = time.perf_counter() - t0
        gc.enable()
        comp_times.append(t_comp)

        print(f"    L Complete [getTop = {pct}%] -> Heap: {t_heap:.4f}s | Competitor: {t_comp:.4f}s",
flush=True)

        self.plot_exp2(percentages, heap_times, comp_times, total_ops)

# --- EXPERIMENT 3 ---
def run_exp3_pop_ratio(self, total_ops=1_000_000, percentages=[0.1, 0.5, 1.0, 5.0, 10.0]):
    print("\n=== Starting Experiment 3: Execution Time vs. pop() Ratio ===", flush=True)
    heap_times, comp_times = [], []

    for pct in percentages:
        print(f"\n[Exp 3] Generating sequence for pop = {pct}% ({total_ops:,} ops)...", flush=True)
        script = self.generator.gen_exp3_sequence(total_ops=total_ops, pop_pct=pct)

        # 1. MaxHeap
        gc.disable()
        heap = MaxHeap()
        t0 = time.perf_counter()
        for op, val in script:

```

```

        if op == 1: heap.push(val)
        elif op == 2: heap.pop()
    t_heap = time.perf_counter() - t0
    gc.enable()
    heap_times.append(t_heap)

    # 2. CompetitorArray (Progress Monitored with Timeout Guard)
    gc.disable()
    comp = CompetitorArray()
    t0 = time.perf_counter()
    timed_out = False

    for idx, (op, val) in enumerate(script, start=1):
        if op == 1: comp.push(val)
        elif op == 2: comp.pop()

        if idx % 250_000 == 0:
            elapsed = time.perf_counter() - t0
            print(f" | Competitor Progress: {idx:}/{total_ops:,} ops ({(idx/total_ops)*100:.0f}%)
| Elapsed: {elapsed:.1f}s", flush=True)
            if elapsed > TIMEOUT_SECONDS:
                print(" L [TERMINATED] Exceeded 3-hour limit (Out of Time)", flush=True)
                comp_times.append("Out of Time")
                timed_out = True
                break

    gc.enable()
    if not timed_out:
        t_comp = time.perf_counter() - t0
        comp_times.append(t_comp)
        print(f" L Complete [pop = {pct}%] -> Heap: {t_heap:.4f}s | Competitor: {t_comp:.4f}s",
flush=True)

    self.plot_exp3(percentages, heap_times, comp_times, total_ops)

# --- EXPERIMENT 4 ---
def run_exp4_heapify_vs_push(self, saved_sequences):
    """Reuses the exact push-only sequences sigma_1 ... sigma_5 generated in Experiment 1."""
    print("\n=== Starting Experiment 4: Batch Construction (Heapify vs Push-One-By-One) ===",
flush=True)
    push_times, heapify_times = [], []
    sizes = list(saved_sequences.keys())

    for N in sizes:
        script = saved_sequences[N] # REUSE exact sequence from Exp 1

        # Method 1: Heapify
        gc.disable()
        heap_build = MaxHeap()
        t0 = time.perf_counter()
        elements = [val for _, val in script]
        heap_build.heapify(elements)
        t_heapify = time.perf_counter() - t0
        gc.enable()
        heapify_times.append(t_heapify)

        # Method 2: Sequential Push
        gc.disable()
        heap_push = MaxHeap()
        t0 = time.perf_counter()
        for _, val in script:
            heap_push.push(val)
        t_push = time.perf_counter() - t0
        gc.enable()
        push_times.append(t_push)

    speedup = t_push / t_heapify if t_heapify > 0 else 0
    print(f" L Complete [N = {N:,}] -> Heapify: {t_heapify:.4f}s | Push: {t_push:.4f}s | Speedup:

```

```

{speedup:.2f}x", flush=True)

    self.plot_exp4(sizes, push_times, heapify_times)

# --- PLOTTING METHODS ---
def plot_exp1(self, sizes, heap_times, comp_times):
    plt.figure(figsize=(9, 5.5))
    plt.plot(sizes, heap_times, marker='o', color='blue', linewidth=2, label='MaxHeap')
    plt.plot(sizes, comp_times, marker='s', color='orange', linewidth=2, label='CompetitorArray')
    plt.title('Experiment 1: Time vs Length of Push-Only Sequence', fontsize=13, fontweight='bold')
    plt.xlabel('Sequence Length (L)', fontsize=11)
    plt.ylabel('Overall Running Time (seconds)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend(fontsize=11)
    plt.tight_layout()
    plt.savefig('experiment1_push_only.png', dpi=300)
    plt.close()

def plot_exp2(self, percentages, heap_times, comp_times, total_ops):
    plt.figure(figsize=(9, 5.5))
    plt.plot(percentages, heap_times, marker='o', color='blue', linewidth=2, label='MaxHeap')
    plt.plot(percentages, comp_times, marker='s', color='orange', linewidth=2, label='CompetitorArray')
    plt.title(f'Experiment 2: Time vs getTop Percentage (L = {total_ops:,})', fontsize=13,
fontweight='bold')
    plt.xlabel('getTop Percentage (%)', fontsize=11)
    plt.ylabel('Overall Running Time (seconds)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend(fontsize=11)
    plt.tight_layout()
    plt.savefig('experiment2_gettop_ratio.png', dpi=300)
    plt.close()

def plot_exp3(self, percentages, heap_times, comp_times, total_ops):
    # Filter out 'Out of Time' string entries for clean numeric plotting
    valid_comp = [t for t in comp_times if isinstance(t, (int, float))]
    valid_pcts = percentages[:len(valid_comp)]

    plt.figure(figsize=(9, 5.5))
    plt.plot(percentages, heap_times, marker='o', color='blue', linewidth=2, label='MaxHeap')
    plt.plot(valid_pcts, valid_comp, marker='s', color='orange', linewidth=2, label='CompetitorArray')
    plt.title(f'Experiment 3: Time vs pop Percentage (L = {total_ops:,})', fontsize=13,
fontweight='bold')
    plt.xlabel('pop Percentage (%)', fontsize=11)
    plt.ylabel('Overall Running Time (seconds)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend(fontsize=11)
    plt.tight_layout()
    plt.savefig('experiment3_pop_ratio.png', dpi=300)
    plt.close()

def plot_exp4(self, sizes, push_times, heapify_times):
    plt.figure(figsize=(9, 5.5))
    plt.plot(sizes, push_times, marker='o', color='purple', linewidth=2, label='push-one-by-one')
    plt.plot(sizes, heapify_times, marker='s', color='green', linewidth=2, label='heapify')
    plt.title('Experiment 4: Heapify vs Push-One-By-One Construction', fontsize=13, fontweight='bold')
    plt.xlabel('Sequence Length (L)', fontsize=11)
    plt.ylabel('Overall Running Time (seconds)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend(fontsize=11)
    plt.tight_layout()
    plt.savefig('experiment4_heapify_vs_push.png', dpi=300)
    plt.close()

```

Appendix E: main_sequential.py

```
"""
main_sequential.py
Main execution for all the four experiments sequentially.
"""

from DataGenerator import DataGenerator
from ExperimentRunner import ExperimentRunner

if __name__ == "__main__":
    generator = DataGenerator()
    runner = ExperimentRunner(generator)

    # 1. Run Exp 1 and capture the 5 push-only sequences sigma_1 ... sigma_5
    saved_sequences = runner.run_exp1_push_only()

    # 2. Run Exp 2
    runner.run_exp2_gettop_ratio()

    # 3. Run Exp 3
    runner.run_exp3_pop_ratio()

    # 4. Run Exp 4 passing the saved sequences from Exp 1
    runner.run_exp4_heapify_vs_push(saved_sequences)

    print("\n=====")
    print(" ALL EXPERIMENTS COMPLETED SUCCESSFULLY!")
    print("=====")
```

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